Despicable Machine

One in a Minion

Table of Contents

Team Introduction & Plan 1-2
Community Outreach 4-6
Design Process 7
Build & CAD 8-10
Code & Sensors 11-14
Season Summary & Future Goals 15

FTC Team

#6200

2024-2025

Our Team

Fun Challenge: We hid minions throughout our portfolio! As you go through, count how many you see, and we'll reveal the anwer on the last page. Good luck!

Bello!

We are Despicable Machine, a 13th year FTC team from Barrington High School.

Mission Statement:

This season, **we strived to create impact on a large scale by inspiring people all over the world to take on STEM and FIRST**, both at a professional and community level. FIRST has become more than just a robot on a field for our team — it's become a way for us to effect meaningful change into the lives of people and to do our part in **creating the next generation of engineers, coders, leaders, and visionaries**. Implementing the principles of "Gracious Professionalism," we aim to learn, share our knowledge, and inspire others to do the same.

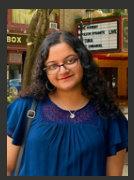
Season Performance Highlights

- **3rd in TeleOp Average** @Meet 3
- **3rd in TeleOp OPR** @Meet 2
- **4th in overall OPR** @Meet 2
- **83 FLL teams mentored**
- **160 robotics kits donated**
- **1st place Control Award**

Specfic Season Goals	How we accomplished them
Raise enough money to fund our outreach and design aspirations	We cold emailed 126 companies, presented to 25, and **gained sponsorships from 18**
Improve the performance of our robot by increasing localization accuracy during autonomous	We implemented unique code such as **4 wheel odometry** and the use of **9 sensors**
Use the design process and custom made parts to enhance the build	We invested in **3D printing materials** and contacted mentors to learn more about design principles
Spread STEM access to groups with limited or no access	We started most of our outreach over the summer, adhering to strict deadlines and focusing on goals

Meet the Team!

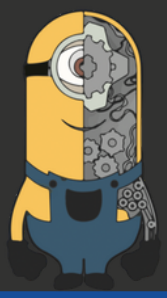
Cindy
Coding, Outreach, Author



Zarah
Notebook, Fundraising

Justin
Building, CAD

Alora
CAD, Notebook



Hafsah
Notebook, Outreach, Author

Virginia
Building, Outreach

Aidan
Fundraising, Coding

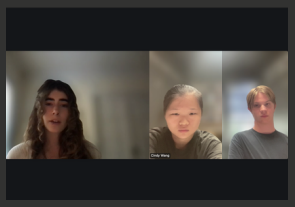
Together, we make Team #6200, **Despicable Machine!** You could say a team like this is ***One in a Minion!***

Team Plan

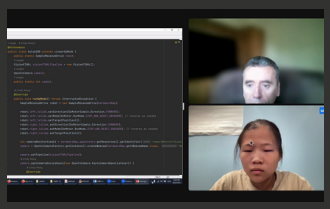
Reaching Out to the Professional Community

This season, we connected with companies and mentors to build our skills and connect with the professional community. We are beyond grateful for the support of our sponsors, which have made our season's activities possible.

Teaching **15 CEOs** about FIRST



Meeting with **Northwestern ESW**



Virtually consulting Dr. Kolmogorov.

Hosting a **workshop at a Microsoft fair**

PIVOTGEN · GENE HAAS FOUNDATION · SWISS AUTOMATION, INC. · Barrington Rotary Club · Coilcraft · Pennino Family Dentistry · MATHNASIUM The Math Learning Center

Recruitment Plan

Mentor Recruitment:

Goal: Recruit a variety of mentors with varying and helpful skill sets (technical and project management)

1. **Cold call/email** >40 STEM professionals
2. Use **connections from outreach events** (e.g. Microsoft STEM fair)
3. Use **LinkedIn** or **personal connections** to contact >15 STEM professionals

→ **16 mentors** recruited by November
- 10 new mentors total

→ **Meet with mentors virtually & in-person**
- Spent ***>45 hours*** meeting with mentors this season

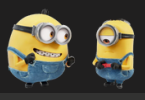
Team Member Recruitment:

Goal: Recruit and train new members to ensure team sustainability and introduce more students to FIRST

Founded a robotics club and Girls Who Code chapter at BHS
- Introduces students to robotics/STEM
- Teaches preliminary skills to prepare for FTC

Promoted FTC recruitment via social media & posters

→ **Interviewed >30 potential recruits**
↳ Decided based on recruits' willingness to commit, display of positive qualities, and dedication to learning

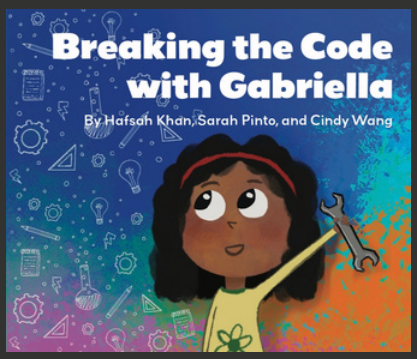


→ **12 new members** recruited to BHS robotics program
- Spread across 3 Barrington teams

Training new members
- **G.E.A.R.S.,** a program designed by our team to teach essential FTC skills to recruits (building, coding, CAD, and portfolio)
- A **one-on-one mentorship program** with new and veteran members

6200 Motivates: Published Our Own Book

This season, the two female co-captains of our team increased the reach of their STEM picture book "Breaking the Code with Gabriella" that they wrote and illustrated last season.



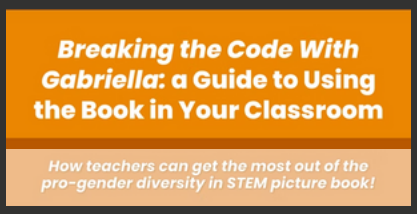
Problem

Low female participation in robotics and gender discrimination in STEM spaces

Our Solution

Writing and illustrating a picture book aimed to inspire young girls to pursue robotics. By creating something aimed for kids, we could spread the values of FIRST to a new age group who can grow up learning to love STEM

- Incorporated our book into **80 schools' elementary curriculum** across 15 districts
- Read our book/promoted FIRST at **9 elementary schools (65 classrooms)**
- Created **Spanish version** for greater reach



Created resources & discussion questions for teachers across ***80 elementary schools!***

Breaking The Code with Gabriella *IMPACT*

- **Part of 80 elementary schools' (15 school districts') elementary STEM curricula: 30,000+ students**
- **One Book One Community, 2500+ books distributed: 18,000+ people (200 families at Winterfest)**
- **ABC 7/Chicago Tribune feature: ~2.2 million people**
- **Read alouds (65 classrooms): 2,000 students**
- **Presence in 13 libraries/new Spanish version: countless future impact!**

One Book One Community

BTCWG chosen for Barrington 220's 1st ever O.B.O.C. initiative!

Every family in the BSD 220 district received a copy of the book, and we discussed its message to 200+ families at Barrington's **Winterfest**.

> **"I was so impressed with the central message of this book... Cindy and Hafsah are role models for all of our young students."**
>
> ***-Dr. Winkelman, Superintendent of 220 Schools***

In the News

We were featured in the **Chicago Tribune** ***(1.7 million daily readers)***, spreading FIRST/gender inclusivity in STEM!

Barrington High School students, from left... More
BARRINGTON SCHOOL DIST. 220

NORTHWEST SUBURBS FOLLOW

Trio of Barrington High School students pen book on their passion for STEM and robotics, call for younger girls to consider it too

MYRNA PETLICKI
October 16 at 6:07 PM CT

We were also featured in a **ABC 7 News segment (500k viewers)**

Stop by our pit to learn more about our book and its impact!

6200 Motivates: Generating Interest

2,443,943 people reached | **280** team hours | **802** man hours | **57** events

Sent 160 Robotics Kits to Underserved Schools

IMPACT: 2,000 students

- Targeted schools **in Zimbabwe** without resources to establish robotics programs
- Used by **thousands of middle schoolers** as their **first exposure to robotics**
- Created opportunities for learning about FIRST and STEM

Created and Distributed 8 Lesson Curriculum

- Beyond providing the resources we **provided instruction**
- Students had **very little to no background** in STEM
- Provided lessons covering robotics basics and further STEM involvement
- With kits, can be **reused annually** for future grade levels **at no cost**

Partner Schools:

- **USAP Community School**
  - Serves high-achieving low-income students and currently has 16 students
- **Uncommon**
  - Provide STEM education to students (mostly first time coders) with 3000+ students

STEM Toy Drive (400+ toys)

IMPACT: 500 students

- Partnered with **4 schools**
- Collected **400+ toys** (estimated value *$6k+*)
- **Donated toys to Alexian Brothers Hospital** and **ICNA Relief Refugee Center**
  - For children hospitalized over the holidays and refugees in underfunded schools

Microsoft STEAM Fair

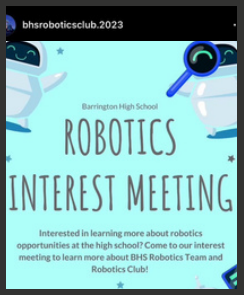
IMPACT: 70 students

- Volunteered at Microsoft Chicago Research Center, **designed & coordinated hands-on robotics and cybersecurity activites**

Robotics Club/Girl Code Club

IMPACT: 70 students

- **Founded/led** Robotics Club & Girl Code; **exposed >70 members to robotics/STEM**

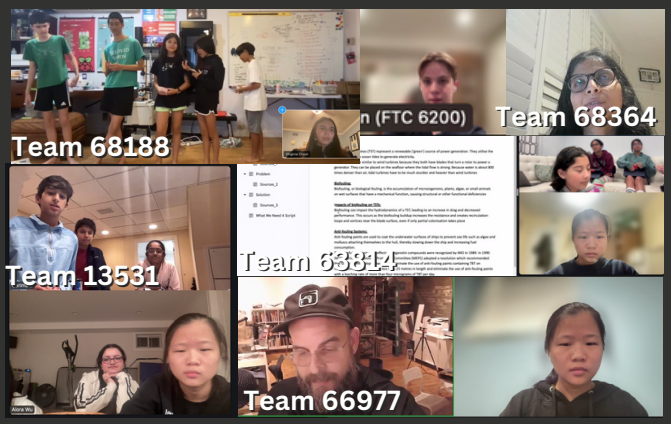


6200 Motivates: Enabling Participation

FLL Mentorship (83 teams)

IMPACT: 750 students

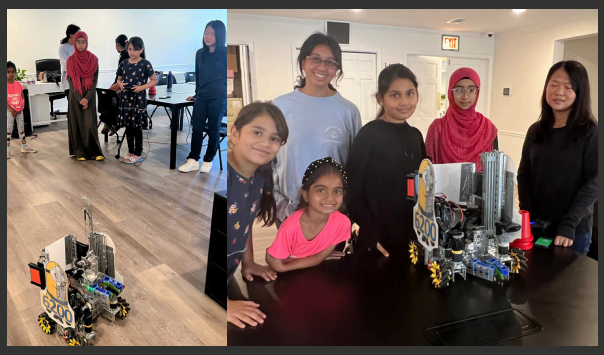
- Hosted **5-week workshop** for **83 FLL teams** from 18 states/regions, 31 being first-time coaches
- Individually **mentored 8 FLL teams in weekly calls** across three months, with **5 winning awards**



Refugee Robotics Camp

IMPACT: 27 students

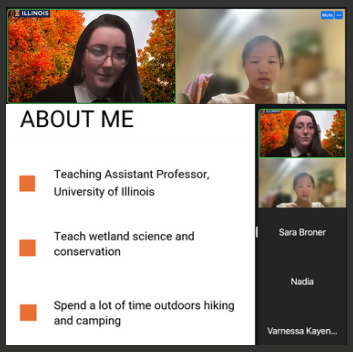
- Started a **weekly STEM camp for refugees** in underfunded schools at ICNA Relief Refugee Center
- Brought our robot and led STEAM activities, teaching them about FIRST and robotics



Gabriella's STEM Challenge

IMPACT: 500 students

- Created a STEM essay competition **to increase robotics interest** in girls in middle/high school
- Hosted **STEM workshops & robotics lessons** with professors from **Northwestern & UIUC** to provide career support



Robotics Showcases

IMPACT: 5,950 people

- Demonstrated our robot interactively to students at:
  - BHS Innovation Fair (250+ students)
  - Oaktoberfest 2024 (300+ people)
  - Mount Prospect FLL Tournament (200+ students)
  - Windy City Bulls Education Day (5,200+ people)
- Spread the values of FIRST & hands-on STEM to thousands of elementary and middle schoolers

"The passion these students have for spreading STEM is beyond inspiring and has really shined through this season." *- Mrs. Dowdy, Elementary School STEM Teacher*

6200 Thinks: Design Process

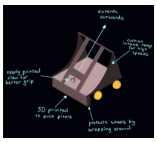
Throughout the season, we used **the design process** to build/improve our robot.

1.Define the Problem

Set specific goals we wanted our robot to achieve:

2.Generate Concepts

Worked together as a team to brainstorm solutions



3.Develop Solution

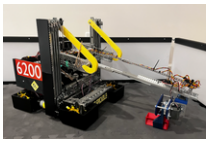
Used Fusion 360 and OnShape to CAD preliminary designs

4.Test Prototype

Created prototypes of best solutions and test on the field

5.Present Final Solution

Build robot using off-the-shelf and custom machined parts



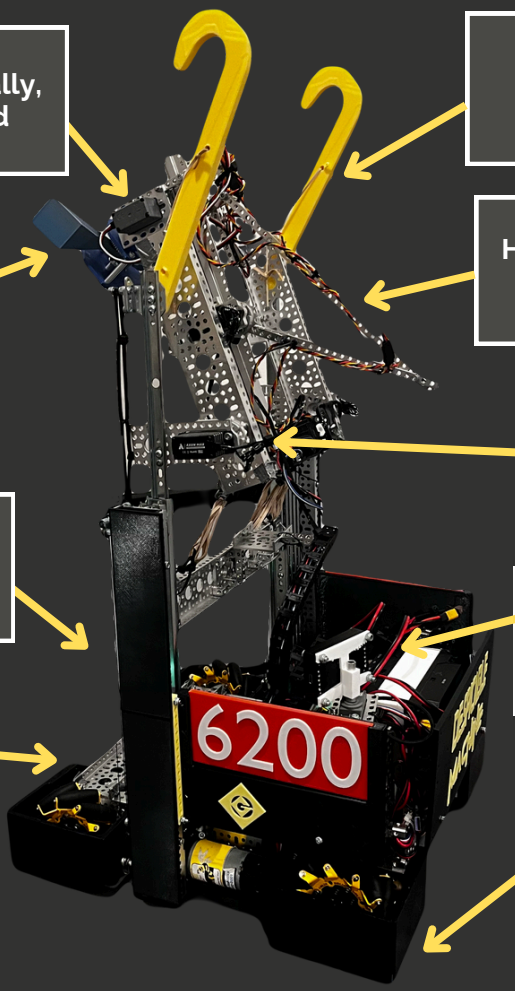
6.Evaluate

Test robot and locate areas of improvement after each meet

- Whole team meetings after each meet to reflect

Dates: 11/14/2024
Attendees: Justin, Cindy, Aidan, Alora, Virginia, Zarah
Goals for Meetings:
- ☐ Reflect on Meet 1
- ☐ Find areas of improvement/plan

An Overview of Our Robot:



Intake
Servo to rotate claw vertically, decrease intake time, and score specimens

Hanging
Spring-loaded hooks tensioned with rubber bands

Extension
Horizontal extension system to reach into submersible and easily score in baskets

Intake/delivery: claw
Custom 3D-printed claw fitted to sample shape

Lift System
Axon servos that rotate claw system 180 degrees to score in the high basket without turning

Delivery
Linear slides that extend 52" to consistently score in the high basket

Vision
Limelight on a servo to rotate camera and increase field of view

Chassis
Modular chassis that allows for rapid iteration

Design
Black wheel guards and side plates to protect our wheels and match our team theme

6200 Designs/Innovates: Our Robot

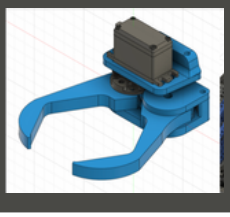
A Deeper Dive Into Our Robot

1 Sample Intake with Claw

GOAL: Pick up samples quickly, even in randomized submersible

Iteration 1

A Loony claw to grab samples horizontally



Thin, dropped samples easily, slow to intake

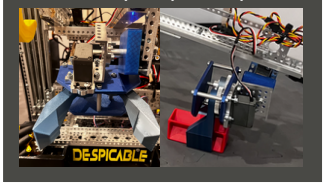
Iteration 2

Custom-designed, thicker claw to match sample shape

Held samples, slow, hard to grab from submersible

Iteration 3

Claw on a servo for horizontal and vertical sample pickup

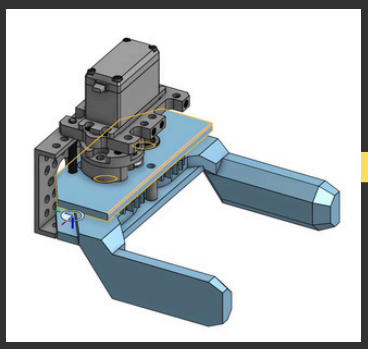


Holds samples, fast, intake anywhere in submersible

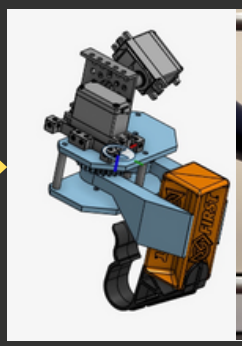
Final Design

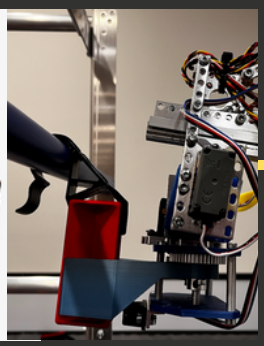
Claw mounted on a servo **rotates vertically and horizontally,** allowing us to pick up colored blocks **quickly** from submersible **without moving blocks** in the way.

Unique Advantage: Specimens, High Basket

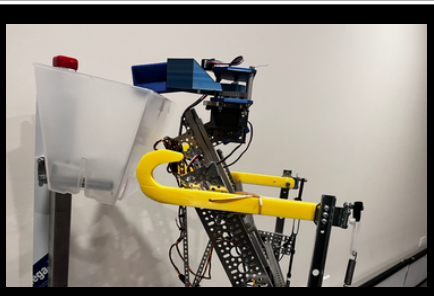


Does not place specimens





Places specimens on high, low chamber

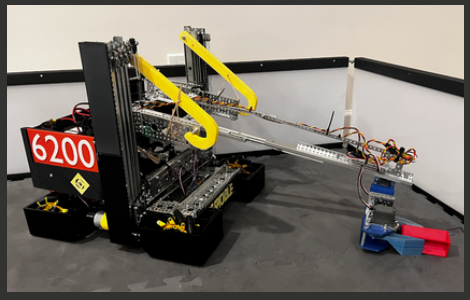


Two Axon servos **rotate the entire claw mechanism 180°**, allowing us to drop samples in the high basket without turning our robot. This **saves us time** throughout the game and helps us score **an additional sample in autonomous.**

Our final design had another **unique benefit**: tackling the scoring of specimens through our rotating claw. The **rotation clips specimens in place**, allowing us to score on the low and high chamber **without any additional mechanism,** following our **design principle of Keeping It Simple Silly (KISS).**

2 Horizontal Extension System

GOAL: Reach and pick up blocks from submersible



Iteration 1: Single low-torque servo → **Iteration 2:** 2 low-torque servos connected by chain → **Iteration 3:** One high-torque servo

Our horizontal extension system allows us to **efficiently and quickly pick up samples** from **deep into the submersible.** It also helps us **adjust our reach to the high basket** without needing to move the whole robot with linear slides extended. We experimented with the above **iteration process,** finding that the singular high torque servo works best to help us **swiftly and accurately extend.**

3 Ascent Mechanism

GOAL: Reach a Level 2 ascent with hang

Problem:

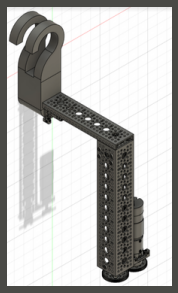
Due to the size of our robot, we aimed to fit our hanging mechanism within the 14x10.25 in chassis without increasing robot weight and slowing it down.

Fusion360 Validation
Use Fusion360 to...

1. Test hooks of different sizes and shapes
2. Find the center of mass and so the hook can be placed over it

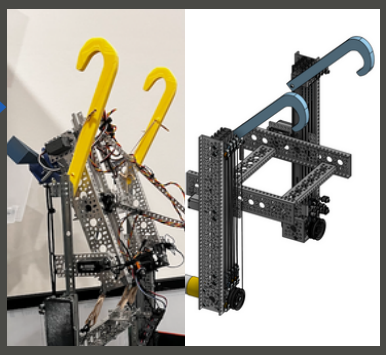
Iteration 1

Wide 3D-printed single hook



Strong grip
Unbalanced, Heavy
Linear actuator hook grips high rung from a single contact point.

Solution:

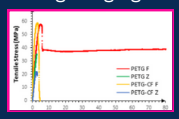


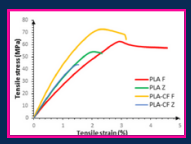
Our final hanging mechanism uses spring-loaded 3D-printed custom hooks tensioned with rubber bands to hold a level 2 ascent.

Testing our Hook

Right: Stress and strain correlation graph

We wanted to ensure that the stress/strain on the hook wouldn't cause it to break. We used a spring scale to run tests on the force exerted on the hook during hanging.



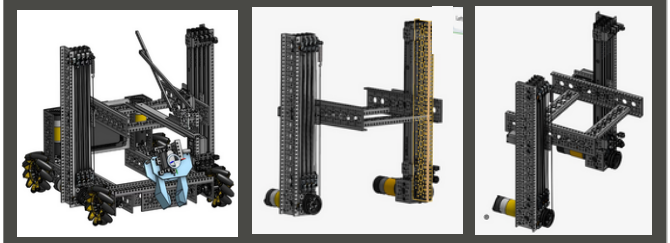


4 Chassis

GOAL: Build a chassis that accommodates for multiple mechanisms

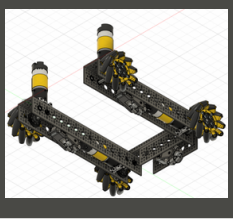
Accomodate linear slides

U-channel for slide support as mechanical backup, small chassis to aid hanging



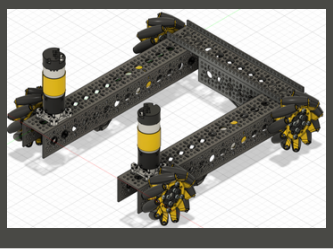
Accomodate Odometry Pods

Vertically mounted motors made space for odo pods



Allow rapid iteration

Open space made developing mechanisms easier



5 Industrial Design

GOAL: Create an efficient robot with design principles

Design Principles

Important concepts we implemented throughout the design process

Keep It Simple Silly (KISS) — **SEE:** Claw, Extension

Used to **simplify our build**, saving time, money, and effort.

Fusion 360 Validation — **SEE:** Hang, Chassis, Design

Developed a **full-robot CAD model in Fusion 360**, allowing us to test specific parts for fit and concept before building them.

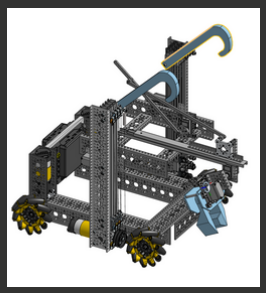
Backed by Calculations — **SEE:** Claw, Hang, Extension

Made decisions **using math and physics to find optimal solutions**, reducing the need for trial and error.

Prototyping and Machining — **SEE:** Design, Claw

Used **custom-made parts to fit our specific design needs.**

We connected with our mentors, such as Mr. Patel of Optimal Design and Dr. Kolmogorov to improve our CAD and design.



*Throughout, we developed meticulous full-robot CAD and used it extensively to **plan out** our building, ensuring we stuck to our budget and met our goals.*

*Our robot **focuses on industrial design principles and functionality,** with wheel covers to protect and give the robot a sleek look.*

Game Strategy

Gru's Gambit: Strategy in Robotics

With so many components to this year's game, we wanted to come up with **focus areas for optimizing development** early on in the season. Examining point values, we found the following maximum points for one robot during each period.

Auto: 68 points (1 specimen, 3 high basket samples)

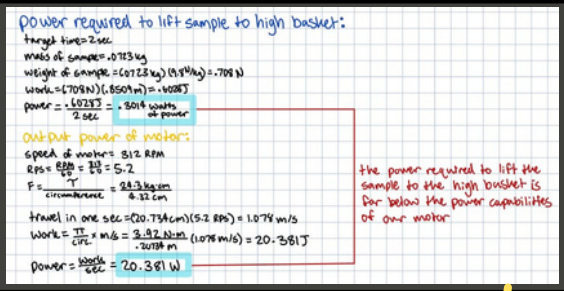
Driver-Control: 80 points (10 high basket samples)

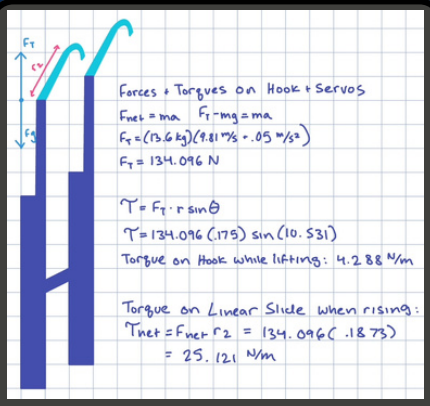
End Game: 15 points

We decided on **consistent driver control and auto with high basket scoring** that helped us **place 3rd** in average TeleOp score in meets 2 and 3.

The Despicable Force: Physics and Strategy in Our Robot

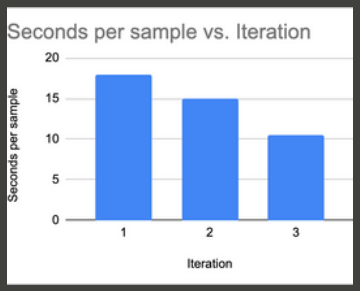
To **save time lifting samples to the high basket**, we calculated the **motor power** required to do so in two seconds and determined our current motor sufficient.

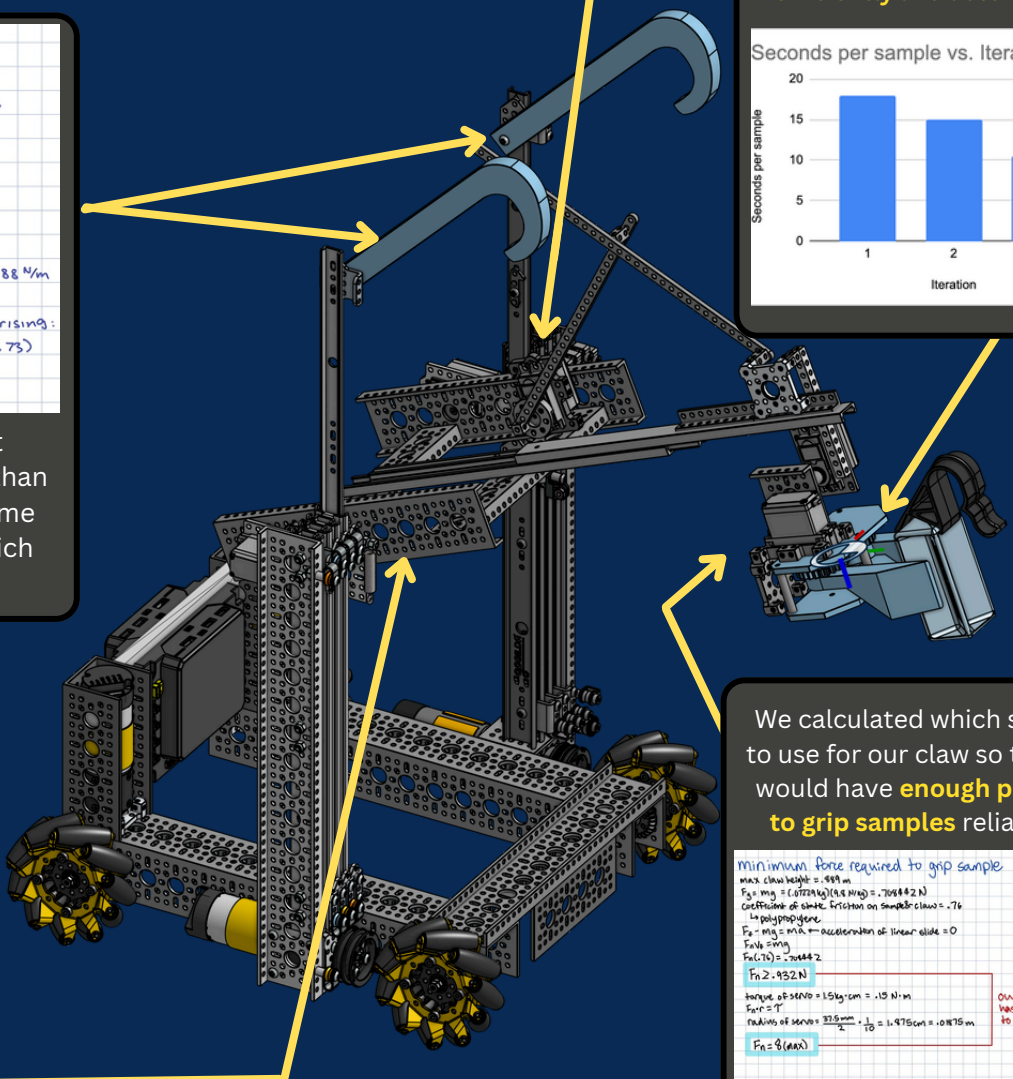


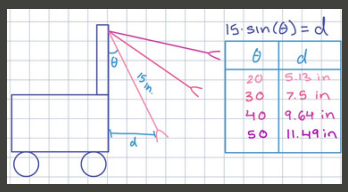


Hanging a heavier robot requires a **stronger motor** than what we had, so we did some **calculations** to decide which motor to purchase.

By incorporating servos to **rotate our claw**, we are able to pick up and place samples in the high buckets more **efficiently and accurately**.

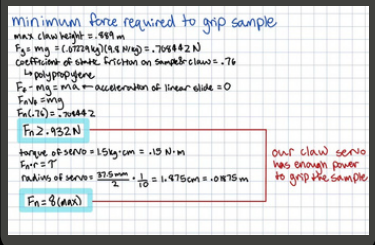






We used **trigonometry** to determine the angle for our **horizontal extension system** that would allow our robot to pick up samples efficiently.

We calculated which servo to use for our claw so that it would have **enough power to grip samples** reliably.



6200 Controls: Our Software

Our Robot Can...

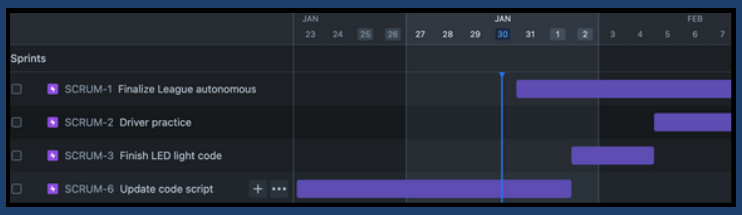
1. Place preloaded specimen on the high chamber **(97% accuracy)**
2. Place three additional samples into the high basket **(95% accuracy)**
3. Use **preprogrammed buttons and sensors** to **assist during driver control**

This season, we tackled our software by setting goals and finding efficient and creative ways to achieve them.

Preliminary Goals/Implementation

GOAL: Give everyone a chance to code

1. Created an **FTC Code Guide** to teach new team members.
2. Used **a Github repository with multiple branches** for version control and to facilitate **collaborative coding**.
3. Utilized **Jira to improve agile workflow**



GOAL: Enhance code workflow

We used git for effective version control throughout the year.

We switched to Android's Wireless ADB for code upload, saving time.

We used a USB Wi-Fi adapter to connect to the Control Hub and use the internet simultaneously.

Motor Control PID Enhancements

Before we started programming autonomous and driver control, we first used software to ensure robot mechanisms worked optimally.

GOAL: Optimize control of linear slides, ensuring they're in sync

At the start of the season...

$$f_1(t) = K_{p1}e_1(t) + K_{i1}\int_0^t e_1(t)dt + K_{d1}\frac{de_1(t)}{dt}$$

$$\neq$$

$$f_2(t) = K_{p2}e_2(t) + K_{i2}\int_0^t e_2(t)dt + K_{d2}\frac{de_2(t)}{dt}$$

...we used the built-in position method.

This varied the speed in each slide & put tension on the rod in between.

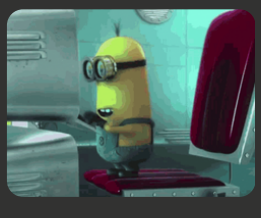
Solution:

$$f(t) = K_p e_{avg}(t) + K_i\int_0^t e_{avg}(t)dt + K_d\frac{de_{avg}(t)}{dt}$$

We developed a **custom PID control algorithm** for the linear slide motors, allowing us to control both slides more effectively.

Autonomous

First, we turned our attention towards autonomous, aiming to get the maximum amount of points possible. To do this, we needed our robot to be highly accurate.



GOAL: Improve accuracy of robot during autonomous

We accomplished this in 4 main ways:

1 Control Loops for Drive Motors

To **minimize error** during autonomous and with automated portions of driver control, we used control loops with proportional, integral, and derivative **(PID)** terms. For our motors, we used the following feedforward control formula to create **custom PID control algorithms.**

$$f(t) = K_v \times velocity + K_a \times acceleration$$

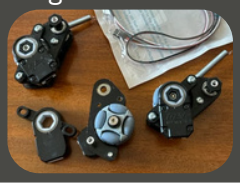
Trapezoidal motion profile for increased accuracy over longer distances

2 Custom 4-Wheel Odometry Design

GOAL: Improve localization accuracy with 4 dead wheels instead of 3

Iteration #1

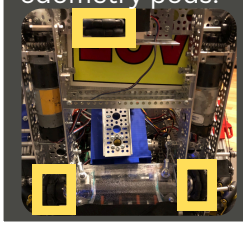
Custom-designed odometry pods with 3 REV Through-Bore Encoders.



Inconsistent ground contact & revolution counting

Iteration #2

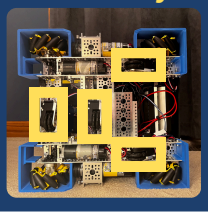
3 goBILDA odometry pods.



*High-resolution encoder gave **precise position information,** Spring loading **improved ground contact***

Iteration #3

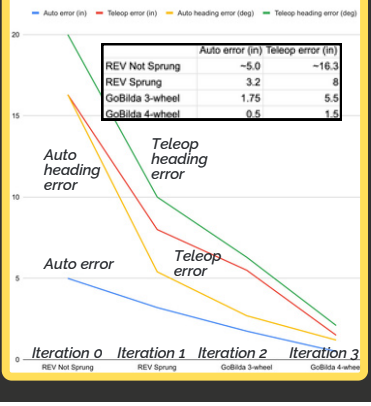
Custom 4-wheel odometry setup.



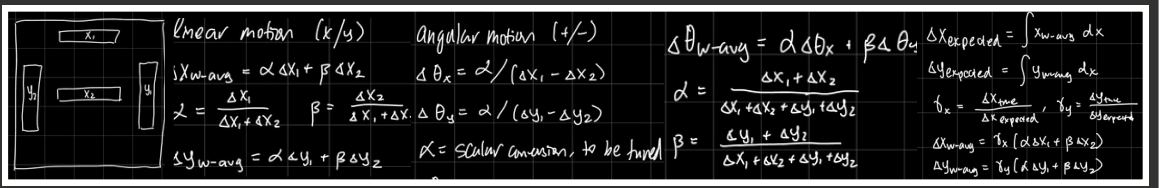
Most teams use only 3 dead wheels.

*Redundant data from the extra wheel **improved localization 3x***

Error by Encoder Iterations



Adaptive Weighted Average Algorithm for 4 encoders



***Takes into account error from encoders**, compares with other localization methods (i.e. Limelight).*

3 Other Enhancements: Time and Setup

Resulting from our improved accuracy, we could use **complex splines** to save time *(right: spline parametric)*

$$x(t) = x_0 + v_x t, y(t) = y_0 + v_y t$$

Async functions move delivery system while maneuvering chassis; saves time

$$dist = 0.206\sqrt{0.0678\,(voltage + 3659.32)} - 3.11, R^2 = 0.99$$

MB1643 ultrasonic sensor for setup positioning

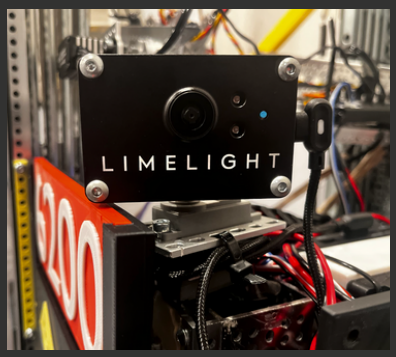
- Checks distance to two walls to get starting position, reduces human error *(right: conversion formula)*

Human error in setup

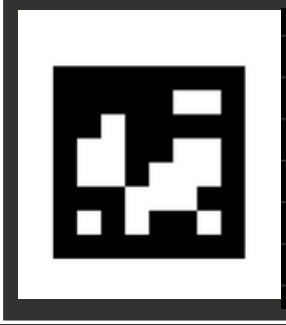
MB1643

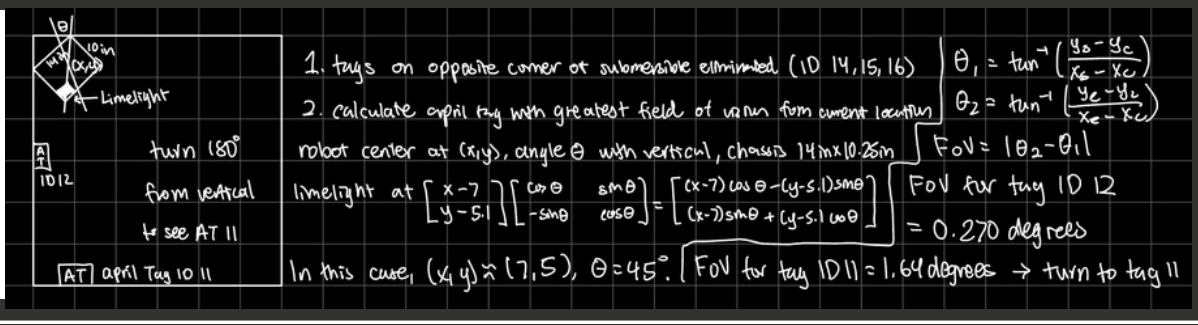
4 Computer Vision: Limelight

GOAL: Localize robot with field AprilTags



We used the **Limelight Smart Camera** which allows us to leverage **the 6 AprilTags**, converting data from the camera into our coordinates on the field and **improving localization.** Placing the camera on a servo allows us to move the Limelight regardless of robot orientation, **increasing the field of view** and allowing us to adjust our autonomous trajectory accordingly.





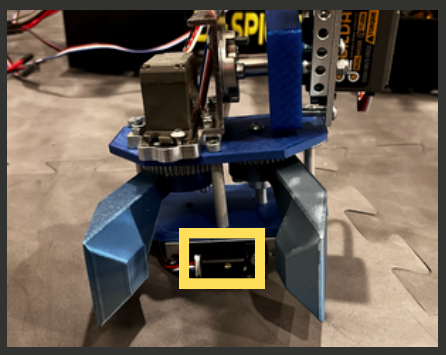
Automation in Driver Control

Next, we focused on using code to make driver control as efficient as possible.

GOAL: Increase automation in driver control to make point scoring easier

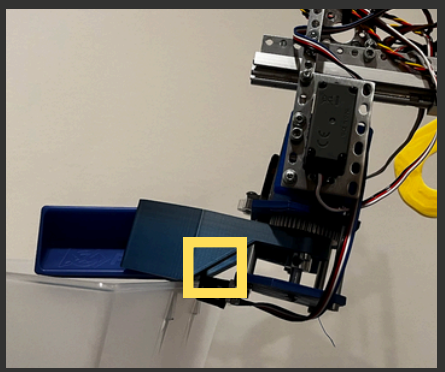
1 Sensor-Assisted Scoring Distance/Intake Detection

Problem: We found that it was often difficult for our drivers to see if the sample was ready to be delivered to the high basket in driver control. We also saw that it was difficult to pick up the correct sample color within the randomized submersible.

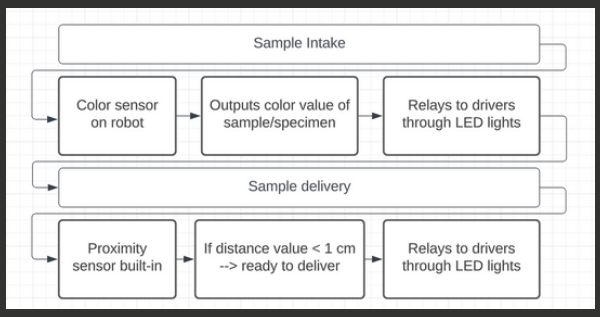


REV V3 Color Sensor and attached **Proximity sensor** → Allows us to **detect color of sample ready to intake**

Combined with **localization and motor encoder data** → Allows us to **automatically detect when in scoring distance of the baskets**

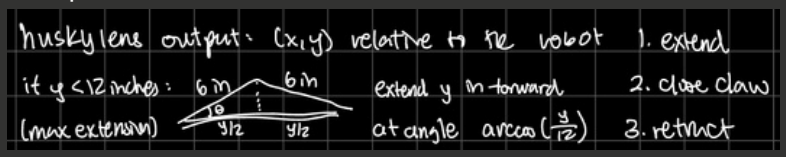


Using the **REV Color Sensor**, we tackled both issues with **a simple solution**, as the built in proximity sensor aids in scoring distance detection to the high basket.



2 HuskyLens Sample Detection

One challenge we saw this year was quickly intaking the correct sample color from the randomized submersible during TeleOp.



To tackle this, we used the **HuskyLens AI Vision Sensor** to **calculate our position relative to samples** and **automatically extend the claw.**

*The HuskyLens' built-in **AI color recognition** helps detect colored samples with almost **100% accuracy** irrespective of minor lighting changes, helping us adapt no matter the field conditions.*

3 Relaying Info Via LED Lights

Huskylens Sample Detection

Holomic Drivetrain
↳ Allows us to move in any direction

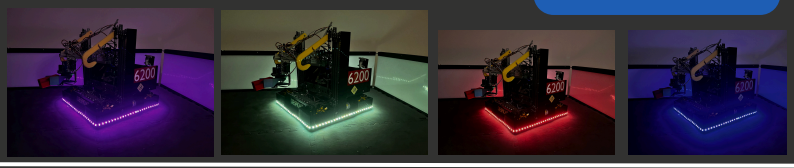
Limelight Vision

REV Color Sensor

Determine if robot is ready to intake and sample color

Detect when robot is within scoring distance of the baskets

Relays that info to the driver via LED lights attached to the robot



LED Color Key

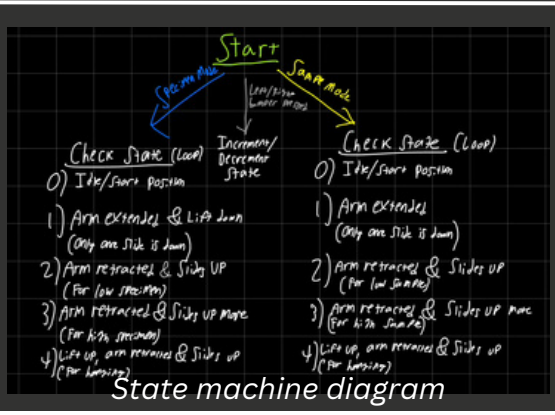
- **Scoring distance**
- **Yellow sample intake**
- **Red sample intake**
- **Blue sample intake**
- **Endgame warnings**

4 State Machines: Decrease Cycle Times

- Automated movement of the arm and claw system.
- Used **position feedback from 6 servos** to **move delivery system to exact spot for scoring with just one button press.**

Other features

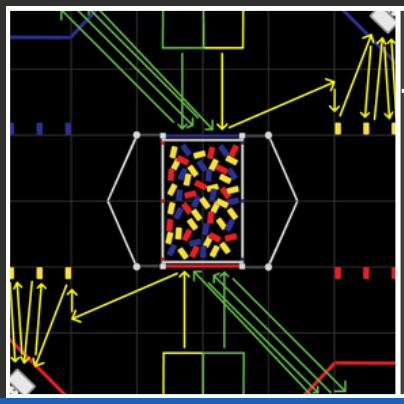
- Control of **extension system for automatic pickup** from submersible, minimizing adjustment time.
- **Specimen and sample modes** for automatic scoring



State machine diagram

6200 Code Summary: *Sensors Used (18) and their Objectives*

ATUO PATH



For Improved Localization	For Motor Control	For Automation in TeleOp
• **1 Limelight** camera for AprilTags detection • **4 GoBILDA odometry pods** for position tracking • **2 MB1643 ultrasonic sensors** for accurate autonomous setup	• **2 motor encoders** on each of the linear slides to track position • **6 pulse width modulators** (PWM) built into each of the servos for controlling movement	• **1 Huskylens** for automated sample detection • **1 REV Color Sensor V3** for TeleOp scoring distance detection • **1 REV Blinkin LED Driver** for LED color control during TeleOp

Team Highlights

- ***MOTIVATE:*** Incorporated STEM/FIRST book into 80 schools, sent robotics kits to 2000 kids, started robotics camp for refugees
- ***CONNECT:*** 16 mentors, teaching us valuable skills
- ***DESIGN/INNOVATE:*** Custom-designed rapid intake/delivery mechanism with extension system, spring-loaded hooks
- ***CONTROL:*** 4 Wheel Odometry/LED light communication system

Lessons Learned

- More **efficient project management,** taught by our mentor **Mrs. Kazmi**
- Started to fix problems as soon as they appear and not procrastinate
- Increased **build efficiency with CAD** and math/physics calculations
- **Secured more sponsorships** with a strong pitch and presentation to companies

Future goals

- Increase reach of the book and continue **expanding STEM curriculum**
- **Further develop GEARS**, the team training program
- Enhance robot performance with new technologies
- **Build strong relationships** with FTC teams in our area

Plans to Achieve Them

- **Engage community members** and new schools to spread FIRST
- Encourage old members to mentor new ones
- **Research machine learning and machining techniques**
- Reach out to nearby teams and **plan collaborative outreach**

Our Total Impact

Impact via Socials @despicablemachine6200

Minion Answer Reveal:
We hid 21 Minions!

We use social media to share our team's journey, connect with other FTC teams, and spread FIRST. This season was the first time our team used social media, and we saw great success, amassing thousands of likes.

Tiktok
168.7k views/**24.7k** likes

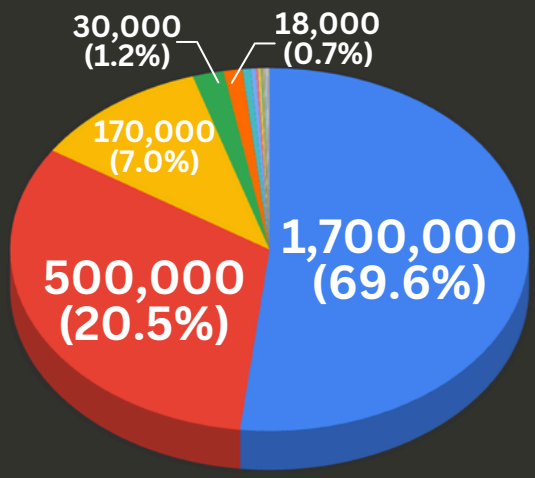
Instagram
639 followers/**1,700** likes

Accounting for overlap between platforms, we've reached
~170k people
through our social media accounts.

People Reached Breakdown

This chart represents the **total number of people our team impacted** through in-person and online events as well as news articles and other features.

SUMMARY
- **802** hours dedicated to outreach
- ...across events **57**



Total:

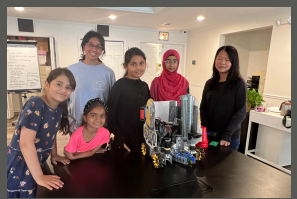
2,443,943 people reached

We are so grateful...

to be able to share our season with you...

to inspire others to take on challenges....

and to collaborate with people around the world!



And if there's one thing we learned this season,
in order to make an impact, someone has to do it...

FIRST®